

Information Criterion

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1 Summary

This document provides a brief overview of Akaike and Bayesian Information Criterion.

2 Assumptions

We have a set $\mathbf{X} = \{x_i\}_{i=1}^N$ of observations from a parametric family of distributions $p(\mathbf{X}|\theta)$ where the true value θ_0 is unknown. We assume that $p(\mathbf{X}|\theta)$ is twice differentiable in θ and $\theta \in \mathbb{R}^k$.

3 Akaike Information Criteria

Let $\hat{\theta}$ be the maximum likelihood estimator of θ and define

$$d(\theta') := \left[\int_{\Omega} -2 \ln p(\mathbf{X}|\theta) d\mathbb{P}(\mathbf{X}|\theta) \right]_{|\theta=\theta'} \quad (1)$$

Then

$$\begin{aligned}
\mathbb{E} \left[d(\hat{\theta}) \right] &= \underbrace{\mathbb{E} \left[-2 \ln p(\mathbf{X}|\hat{\theta}) \right]}_A \\
&\quad + \underbrace{\mathbb{E} \left[2 \ln p(\mathbf{X}|\hat{\theta}) - 2 \ln p(\mathbf{X}|\theta_0) \right]}_B \\
&\quad + \underbrace{\mathbb{E} \left[d(\hat{\theta}) + 2 \ln p(\mathbf{X}|\theta_0) \right]}_C
\end{aligned} \tag{2}$$

Using a second order Taylor expansion around $\hat{\theta}$,

$$\begin{aligned}
B &= 2\mathbb{E} \left[\ln p(\mathbf{X}|\hat{\theta}) - \left\{ \ln p(\mathbf{X}|\hat{\theta}) + \frac{\partial}{\partial \theta} \ln p(\mathbf{X}|\theta) \Big|_{\theta=\hat{\theta}} (\theta_0 - \hat{\theta}) \right\} \right] \\
&\quad - 2\mathbb{E} \left[\frac{1}{2} (\theta_0 - \hat{\theta})' \frac{\partial^2}{\partial \theta \partial \theta'} \ln p(\mathbf{X}|\theta) \Big|_{\theta=\hat{\theta}} (\theta_0 - \hat{\theta}) + o_p(1) \right] \\
&= 2\mathbb{E} \left[\frac{1}{2} (\theta_0 - \hat{\theta})' \frac{-\partial^2}{\partial \theta \partial \theta'} \ln p(\mathbf{X}|\theta) \Big|_{\theta=\hat{\theta}} (\theta_0 - \hat{\theta}) + o_p(1) \right]
\end{aligned} \tag{3}$$

The term $(\theta_0 - \hat{\theta})' \frac{-\partial^2}{\partial \theta \partial \theta'} \ln p(\mathbf{X}|\theta) \Big|_{\theta=\hat{\theta}} (\theta_0 - \hat{\theta})$ converges asymptotically to χ_k^2 by the Central Limit Theorem and the Continuous Mapping Theorem. Therefore $B \rightarrow_p k$.

Likewise, using a second order Taylor expansion around θ_0 ,

$$\begin{aligned}
C &= \mathbb{E} \left[d(\theta_0) + \frac{\partial}{\partial \theta} d(\theta) \Big|_{\theta=\theta_0} (\hat{\theta} - \theta_0) \right] \\
&\quad + \mathbb{E} \left[\frac{1}{2} (\hat{\theta} - \theta_0)' \frac{\partial^2}{\partial \theta \partial \theta'} d(\theta) \Big|_{\theta=\theta_0} (\hat{\theta} - \theta_0) + o_p(1) + 2 \ln q(\mathbf{X}|\theta_0) \right] \\
&= \mathbb{E} \left[\frac{1}{2} (\hat{\theta} - \theta_0)' \frac{\partial^2}{\partial \theta \partial \theta'} d(\theta) \Big|_{\theta=\theta_0} (\hat{\theta} - \theta_0) + o_p(1) \right] \\
&= \mathbb{E} \left[(\hat{\theta} - \theta_0)' \frac{-\partial^2}{\partial \theta \partial \theta'} \ln p(\mathbf{X}|\theta) \Big|_{\theta=\theta_0} (\hat{\theta} - \theta_0) + o_p(1) \right]
\end{aligned} \tag{4}$$

Because $(\theta_0 - \hat{\theta})' \frac{-\partial^2}{\partial \theta \partial \theta'} \ln p(\mathbf{X}|\theta) \Big|_{\theta=\theta_0} (\theta_0 - \hat{\theta})$ converges asymptotically to χ_k^2 , $C \rightarrow_p k$.

It follows that

$$\mathbb{E} \left[d \left(\hat{\theta} \right) \right] = \mathbb{E} \left[-2 \ln p(\mathbf{X}|\hat{\theta}) \right] + 2k + o(1) \quad (5)$$

It follows that the Akaike Information Criterion,

$$AIC := 2k - 2 \ln p(\mathbf{X}|\hat{\theta}) \quad (6)$$

satisfies

$$\begin{aligned} \mathbb{E} [AIC] &= 2k + 2\mathbb{E} \left[-\ln p(\mathbf{X}|\hat{\theta}) \right] \\ &= \mathbb{E} \left[d \left(\hat{\theta} \right) \right] + o(1) \end{aligned} \quad (7)$$

4 Bayesian Information Criteria

Let $p(\theta)$ be a prior on θ . Then using a second order Taylor expansion around $\hat{\theta}$ and Laplace's Approximation, we have

$$\begin{aligned} p(\mathbf{X}) &= \int_{\Theta} p(\mathbf{X}|\theta)p(\theta)d\theta \\ &= \int_{\Theta} \exp [\ln p(\mathbf{X}|\theta)]p(\theta)d\theta \\ &= \int_{\Theta} \exp \left[\ln p(\mathbf{X}|\hat{\theta}) + \frac{\partial}{\partial \theta} \ln p(\mathbf{X}|\theta)|_{\theta=\hat{\theta}}(\theta - \hat{\theta}) \right] p(\theta)d\theta \\ &\quad + \int_{\Theta} \exp \left[\frac{1}{2}(\theta - \hat{\theta})' \frac{\partial^2}{\partial \theta \partial \theta'} \ln p(\mathbf{X}|\theta)|_{\theta=\hat{\theta}}(\theta - \hat{\theta}) + o_p(1) \right] p(\theta)d\theta \\ &\approx \int_{\Theta} p(\mathbf{X}|\hat{\theta}) \exp \left[\frac{n}{2}(\theta - \hat{\theta})' \frac{\partial^2}{\partial \theta \partial \theta'} \frac{1}{n} \ln p(\mathbf{X}|\theta)|_{\theta=\hat{\theta}}(\theta - \hat{\theta}) \right] \exp [o_p(1)]p(\hat{\theta})d\theta \\ &\approx p(\mathbf{X}|\hat{\theta}) \left(\frac{2\pi}{n} \right)^{k/2} \left[\det \left(\frac{-\partial^2}{\partial \theta \partial \theta'} \frac{1}{n} \ln p(\mathbf{X}|\theta)|_{\theta=\hat{\theta}} \right) \right]^{-1/2} p(\hat{\theta}) \end{aligned} \quad (8)$$

It follows that

$$\begin{aligned}
\ln p(\mathbf{X}) &\approx \ln p(\mathbf{X}|\hat{\theta}) + \frac{k}{2} \ln 2\pi - \frac{k}{2} \ln n \\
&\quad - \frac{1}{2} \ln \left[\det \left(\frac{-\partial^2}{\partial \theta \partial \theta'} \frac{1}{n} \ln p(\mathbf{X}|\theta) \Big|_{\theta=\hat{\theta}} \right) \right] + \ln p(\hat{\theta}) \\
&= \ln p(\mathbf{X}|\hat{\theta}) - \frac{k}{2} \ln n + O_p(1)
\end{aligned} \tag{9}$$

so that

$$\begin{aligned}
BIC &:= k \ln n - 2 \ln p(\mathbf{X}|\hat{\theta}) \\
&\approx -2 \ln p(\mathbf{X}) + O_p(1)
\end{aligned} \tag{10}$$